

Hyperbolicity Cones Without Semidefinite Lifts

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Abstract

We prove that not every hyperbolicity cone is a spectrahedral shadow: some closed hyperbolicity cones admit no finite affine semidefinite lift, even with arbitrary real coefficients and any finite number of auxiliary variables. This disproves the Projected Lax Conjecture and hence the generalized Lax conjecture.

1 Introduction

A real homogeneous polynomial p on a finite-dimensional real vector space is *hyperbolic with respect to e* if $p(e) > 0$ and, for every x , all roots of $t \mapsto p(te - x)$ are real. Its closed hyperbolicity cone is

$$K(p, e) = \{x : \text{every root of } t \mapsto p(te - x) \text{ is nonnegative}\}.$$

Gårding proved that this is a convex cone [4]. The determinant on real symmetric matrices, with direction the identity, gives the positive semidefinite cone as a basic example. Hyperbolicity cones also support logarithmic self-concordant barriers, as shown by Güler [6], and hence form a natural setting for convex optimization beyond semidefinite programming.

A *spectrahedron* is a set defined by a finite affine linear matrix inequality. A *spectrahedral shadow*, or a set with a finite semidefinite lift, is a set $S \subset \mathbb{R}^d$ of the form

$$S = \left\{ x \in \mathbb{R}^d : \exists u \in \mathbb{R}^a, A_0 + \sum_{i=1}^d x_i A_i + \sum_{j=1}^a u_j B_j \succeq 0 \right\}, \quad (1.1)$$

where a and the common size of the real symmetric matrices are finite. Equality in this definition is exact; taking the closure of the displayed projection is not part of the definition. The *generalized Lax conjecture* asserts that every hyperbolicity cone is a spectrahedron. The weaker *Projected Lax Conjecture*, formulated by Netzer and Sanyal [17], asks whether every hyperbolicity cone is a spectrahedral shadow.

Theorem 1.1. *There exists a real homogeneous hyperbolic polynomial whose closed hyperbolicity cone has no finite semidefinite lift.*

Thus both conjectures have a negative resolution. The assertion excludes every finite choice in (1.1), with arbitrary real coefficients and any number of auxiliary variables. The construction is an existence argument in high dimension, not a lower bound on the size of a particular representation.

1.1 Context and earlier obstructions

The original Lax question concerned homogeneous polynomials in three variables [13]. The determinantal theorem of Helton and Vinnikov [8], together with the equivalence explained by Lewis, Parrilo, and Ramana [15], gave an affirmative answer: a ternary hyperbolic polynomial has a definite symmetric determinantal representation. The distinction between a representation of a polynomial and a representation of its cone becomes essential in higher dimension. Brändén [2] constructed a real-zero polynomial for which no positive integral power has a definite determinantal representation. This does not exclude another polynomial with the same positivity region. Indeed, Kummer [9] proved that the related specialized Vámos polynomial has a spectrahedral hyperbolicity cone. Nor does failure of a specified sums-of-squares relaxation, such as those studied by Kummer, Plaumann, and Vinzant [11], exclude all finite lifts.

Positive representation results cover substantial classes. Brändén [3] proved spectrahedrality for elementary symmetric hyperbolicity cones. Building on the representability criteria of Helton and Nie [7], Netzer and Sanyal [17, Theorem 1.1] proved that a hyperbolicity cone is a spectrahedral shadow when each nonzero boundary point is a smooth point of its defining hyperbolic polynomial. Scheiderer [19] subsequently obtained second-order-cone lifts for hyperbolicity cones with Nash-smooth boundary. Recent preprints prove spectrahedrality for five-variable hyperbolic cubics [16] and for strictly hyperbolic polynomials [10]. These results concern special families or impose regularity hypotheses; they do not yield an exact finite lift for every hyperbolicity cone.

For general convex semialgebraic sets, Scheiderer [18] established nonexistence of semidefinite lifts by local positivity obstructions. His finite-dimensional sum-of-squares pullback criterion and local arguments explain how a putative lift constrains functions near a smooth point. He explicitly asked whether these methods could settle the hyperbolicity-cone question [18, Section 5.4]. The argument here is closely related to that approach, but tests analytic Gram identities directly by a finite positive moment functional. Moment and sum-of-squares semidefinite methods are classical [12]; our functional need not be integration against a measure. The general relationship between cone lifts and positive factorizations is developed by Gouveia, Parrilo, and Thomas [5]. We do not need to invoke their factorization theorem.

1.2 The construction and proof strategy

We begin with a matrix-valued homogeneous quadratic map $Q : \mathbb{R}^m \rightarrow \text{Sym}_r$ that is positive semidefinite at every real input. Here Sym_r denotes the real symmetric r -by- r matrices. Section 2 associates to Q a homogeneous polynomial p_Q and proves its hyperbolicity by a symmetric determinant identity on each line in the input space. It also identifies an exact affine section and projection of its closed cone:

$$E_Q = \{(A, t, y) : A \succeq 0, \text{ran } Q(y) \subseteq \text{ran } A, t \geq \text{tr}(A^\dagger Q(y))\}.$$

The symbol $A^\dagger$ denotes the Moore–Penrose inverse. It is enough to choose Q so that E_Q has no finite lift.

The difficulty is to force such an obstruction while keeping $Q(y)$ positive semidefinite for *every* y . Sections 3 and 4 address this constraint. Let H be the 36-dimensional space of quadratic forms in eight variables. Each linear functional y on quartic forms determines a Hankel matrix $Y(y)$ on H by $Y(y)(f, g) = y(fg)$. We construct a rank-twenty positive semidefinite Hankel matrix whose kernel has strong multiplication properties. After normalizing translations and dilations, its geometry yields a twenty-dimensional subspace $W \subset H$ and a finite set of projective classes of nonzero symmetric forms on W . The class of every nonzero limit of inverses of compressions to

W arising in the separation argument belongs to this set. The seed matrix is established by the finite-field certificate in Appendix A; the passage from its rank computations to a real positive semidefinite matrix is part of the proof.

For a sufficiently large integer N , finiteness lets us choose a tuple $w = (w_1, \dots, w_N) \in W^N$ with $\sum_i w_i^T C w_i \neq 0$ for every representative C of one of these classes. We then choose $U \subset w^\perp \subset H^N$ so that convergence to w of vectors $(I_N \otimes Y(y))u$, $u \in U$, would force one of these pairings to vanish. Thus w lies outside the closure of those Hankel images. A symmetric form g can be chosen nonnegative on every such image and negative at w . Using a basis matrix also denoted by U , we set

$$G(y) = I_N \otimes Y(y), \quad Q(y) = U^\top G(y) g G(y) U.$$

Nonnegativity on the Hankel images proves that Q is admissible. The construction uses $m = 330$ and $r = 20N - 1$, where one may take $N > 10 \cdot 36^2$.

Section 5 turns the negative direction of g into a local obstruction. Evaluation functionals at points $(1, s)$, $s \in \mathbb{R}^7$, give a degree-four polynomial patch $x(s)$ in E_Q . A positive definite moment functional Λ , applied to $x(c + \varepsilon s)$, produces points violating the trace inequality of E_Q by a negative term of order ε^4 . At the same time their A coordinates are bounded below by a positive multiple of $\varepsilon^4 I$.

If a finite lift existed, semialgebraic selection would provide analytic lift coordinates and analytic Gram factors on an open part of this patch. Applying Λ to their Taylor polynomials almost preserves the lift inequality. A mixing argument on the common support of the feasible pencil values repairs the small error without a strict feasibility assumption. It yields feasible points within $O(\varepsilon^{16})$ of the tested patch, too close to remove its order-four violation. The finite positive extension and finite-jet approximation lemmas isolate reusable parts of this obstruction. Together with the finite inverse-limit construction, they supply the proof of Theorem 1.1.

2 A determinant construction and its affine section

The first step is to place the obstruction in a tractable affine section. We prove hyperbolicity for every admissible quadratic matrix and identify the section exactly, including its singular boundary.

Let $Q : \mathbb{R}^m \rightarrow \text{Sym}_r$ have homogeneous quadratic entries and satisfy $Q(y) \succeq 0$ for every y . Put $n = r + 1$. For $W = \begin{pmatrix} a & v^\top \\ v & T \end{pmatrix} \in \text{Sym}_n$, define

$$\Phi_y(W) = \begin{pmatrix} \text{tr}(Q(y)T) & -v^\top Q(y) \\ -Q(y)v & aQ(y) \end{pmatrix}. \quad (2.1)$$

The polynomial used throughout the paper is

$$p_Q(X, Z, y) = \det((\det X)Z - \Phi_y(\text{adj } X)), \quad X, Z \in \text{Sym}_n, \quad y \in \mathbb{R}^m. \quad (2.2)$$

The adjugate in this formula is the classical adjugate, so the formula is polynomial even when X is singular.

Proposition 2.1. *The polynomial p_Q is homogeneous of degree $n(n+1)$ and hyperbolic with respect to $e = (I_n, I_n, 0)$, with $p_Q(e) = 1$. Fix y and choose vectors $b_1, \dots, b_n \in \mathbb{R}^r$ with $Q(y) = \sum_{j=1}^n b_j b_j^\top$. Define*

$$B_j = \begin{pmatrix} 0 & b_j^\top \\ -b_j & 0 \end{pmatrix}, \quad B = \begin{pmatrix} B_1 \\ \vdots \\ B_n \end{pmatrix}.$$

Then the closed hyperbolicity cone K_Q of p_Q is characterized, at this value of y , by

$$(X, Z, y) \in K_Q \iff \begin{pmatrix} I_n \otimes X & \mathcal{B} \\ \mathcal{B}^\top & Z \end{pmatrix} \succeq 0. \quad (2.3)$$

Proof. The entries of the matrix in (2.2) have degree $n + 1$, and $p_Q(e) = 1$, which proves the degree assertion. A Gram factorization as in the statement exists since $n > r$; zero vectors may be added. Direct multiplication gives

$$\Phi_y(W) = \sum_{j=1}^n B_j^\top W B_j.$$

For every real η we have the polynomial identity

$$p_Q(X, Z, \eta y) = \det \begin{pmatrix} I_n \otimes X & \eta \mathcal{B} \\ \eta \mathcal{B}^\top & Z \end{pmatrix}. \quad (2.4)$$

Indeed, when X is invertible the right side is

$$(\det X)^n \det(Z - \eta^2 \Phi_y(X^{-1})),$$

which equals the left side by homogeneity of Q and the adjugate identity. Equality everywhere follows because both sides are polynomials in X, Z, η .

Write $D_y(X, Z)$ for the symmetric matrix on the right of (2.3). Substituting $tI_n - X, tI_n - Z, -y$ in (2.4) gives

$$p_Q(te - (X, Z, y)) = \det(tI_{n(n+1)} - D_y(X, Z)).$$

Its roots are real, being the eigenvalues of a real symmetric matrix. They are all nonnegative exactly when $D_y(X, Z) \succeq 0$, proving both hyperbolicity and (2.3). $\square$

The Gram factorization in Proposition 2.1 is chosen separately for each y . In particular, (2.3) is not asserted to be a linear matrix inequality jointly in X, Z, y . The closed cone K_Q is convex by Gårding's theorem [4].

For a positive semidefinite matrix A , let $A^\dagger$ denote its Moore–Penrose inverse. Define

$$E_Q = \left\{ (A, t, y) : \begin{array}{l} A \in \text{Sym}_r, \ A \succeq 0, \quad \text{ran } Q(y) \subseteq \text{ran } A, \\ t \geq \text{tr}(A^\dagger Q(y)) \end{array} \right\}. \quad (2.5)$$

Proposition 2.2. *The set E_Q is the image of the affine section $K_Q \cap \{X = \text{diag}(1, A) : A \in \text{Sym}_r\}$ under the projection $(X, Z, y) \mapsto (A, Z_{00}, y)$. Consequently, if K_Q is a spectrahedral shadow, then so is E_Q .*

Proof. For a symmetric block matrix $\begin{pmatrix} H & C \\ C^\top & Z \end{pmatrix}$, positive semidefiniteness is equivalent to

$$H \succeq 0, \quad \text{ran } C \subseteq \text{ran } H, \quad Z - C^\top H^\dagger C \succeq 0.$$

To see the boundary assertion directly, positivity forces every vector in $\ker H$ to annihilate C , by testing the quadratic form on $(u, \lambda v)$. The asserted range condition then follows, and a triangular congruence reduces the matrix to the direct sum of H and $Z - C^\top H^\dagger C$.

Apply this criterion to (2.3) with $X = \text{diag}(1, A)$. Its range condition says $b_j \in \text{ran } A$ for every j , equivalently $\text{ran } Q(y) \subseteq \text{ran } A$. Moreover,

$$\mathcal{B}^\top (I_n \otimes X)^\dagger \mathcal{B} = \Phi_y(\text{diag}(1, A^\dagger)) = \text{diag}(\text{tr}(A^\dagger Q(y)), Q(y)).$$

It follows that every point of the affine section projects into E_Q . Conversely, given $(A, t, y) \in E_Q$, take $Z = \text{diag}(t, Q(y))$ to obtain a point of that section. This proves the asserted equality with no closure operation. Affine sections and coordinate projections preserve the existence of an affine semidefinite lift, proving the final assertion. $\square$

We will construct Q for which E_Q has no such lift. The rest of the argument therefore concerns (2.5), not the much larger determinant in (2.2).

3 Hankel matrices and finitely many inverse limits

We construct a positive semidefinite Hankel matrix of rank 20 and a 20-dimensional subspace W with a finiteness property: certain nonzero limits of inverses of compressions to W have only finitely many projective directions. The construction has two parts. A rank calculation, proved in Appendix A, supplies a point at which a kernel-projection map has sufficiently large derivative. We then divide out translations and dilations, choose a finite fiber of the resulting map, and use that fiber to classify the inverse limits.

3.1 The rank stratum and the seed matrix

Let $x = (x_0, x_1, \dots, x_7)$ and put

$$H = \mathbb{R}[x]_2 = L \oplus P, \quad L = \langle x_0^2, x_0x_1, \dots, x_0x_7 \rangle, \quad P = \mathbb{R}[x_1, \dots, x_7]_2.$$

Subscripts denote homogeneous degree. Monomials are our coordinate bases, and these bases are orthonormal whenever we take an orthogonal complement or compression. Setting $x_0 = 1$ identifies H with polynomials of degree at most two in $s = (s_1, \dots, s_7)$. In this identification, L consists of the constant and linear terms and P consists of the quadratic terms. Write

$$h = \dim H = 36, \quad d_* = 20, \quad k = h - d_* = 16, \quad q = d_* - \dim L = 12. \quad (3.1)$$

Let $\mathcal{H} \subset \text{Sym}(H)$ be the space of Hankel matrices

$$Y(p, p') = y(pp'), \quad p, p' \in H, \quad y \in \mathbb{R}[x]_4^*. \quad (3.2)$$

Multiplication $\text{Sym}^2 H \rightarrow \mathbb{R}[x]_4$ is onto, so $y \mapsto Y$ is injective and $\dim \mathcal{H} = \binom{11}{4} = 330$. We also call Y a moment matrix. This terminology refers only to (3.2); no representing measure is assumed. In the chart $x_0 = 1$, the functional y acts on polynomials in s of degree at most four.

For subspaces of polynomial spaces, a product such as JH_0 denotes the linear span of all products of one element of each space. We use the following algebraic input.

Lemma 3.1 (Seed matrix). *There is a positive semidefinite matrix $T \in \mathcal{H}$ of rank $d_* = 20$ with the following properties. Its restriction to L is positive definite. If*

$$J = \ker T, \quad K = \text{proj}_P J, \quad H_0 = \{p \in H : T(p, L) = 0\},$$

then $\dim K = k = 16$, multiplication

$$\text{Sym}^2 K \longrightarrow \mathbb{R}[x_1, \dots, x_7]_4$$

is injective, and

$$K \langle x_1, \dots, x_7 \rangle = \mathbb{R}[x_1, \dots, x_7]_3. \quad (3.3)$$

Finally, JH_0 has codimension 9 in $\mathbb{R}[x]_4$.

The proof in Appendix A begins with two cyclic orbits of real points and deforms their moment matrix inside the rank-20 stratum. All rank conditions needed for that deformation are verified there by explicit minors.

Let $\mathcal{X}$ be the set of projective classes $[Y] \in \mathbf{P}(\mathcal{H})$ such that $\text{rank } Y = d_*$, $\ker Y \cap L = 0$, and the projected kernel $K_Y = \text{proj}_P(\ker Y)$ satisfies the two multiplication conditions of Lemma 3.1. Positivity is not part of the definition of $\mathcal{X}$. Define

$$\pi : \mathcal{X} \longrightarrow \text{Gr}(k, P), \quad \pi([Y]) = K_Y.$$

Lemma 3.2. *The set $\mathcal{X}$ is a smooth semialgebraic manifold of dimension*

$$D = 330 - 1 - \frac{k(k+1)}{2} = 193.$$

Before projectivization, the tangent space at a representative Y is the space of $Y' \in \mathcal{H}$ whose restriction to $\ker Y \times \ker Y$ is zero.

Proof. Put $J_Y = \ker Y$. Taking highest homogeneous parts in the affine chart identifies J_Y with K_Y . Injectivity of multiplication on $\text{Sym}^2 K_Y$ therefore implies injectivity on $\text{Sym}^2 J_Y$: a relation among products in J_Y would have a relation among their highest parts. By duality, restriction of Hankel forms gives a surjection

$$\mathcal{H} \longrightarrow \text{Sym}(J_Y).$$

Choose a complement of J_Y in H . The compression of Y to this complement is invertible: a vector in the complement orthogonal for Y to that complement is orthogonal for Y to all of H , and hence lies in J_Y . In a neighborhood of Y , rank d_* is thus equivalent to the vanishing of a k -by- k symmetric Schur complement. At Y , the derivative of this Schur complement is restriction to J_Y . The implicit function theorem gives a smooth manifold of codimension $k(k+1)/2 = 136$ in $\mathcal{H}$, with the asserted tangent space. The condition $\ker Y \cap L = 0$ and the multiplication conditions on K_Y are open rank conditions. Finally, projectivization removes the nonzero scalar parameter, giving dimension 193. $\square$

3.2 Removing translations and dilations

The fibers of π contain the orbits of affine translations and positive dilations in the s variables. We give explicit coordinates for removing these eight parameters. Because $J_Y \cap L = 0$, its elements, in the affine chart, have a unique description

$$J_Y = \{p(s) + b(p)(s) + a(p) : p \in K_Y\}, \quad (3.4)$$

where $b : K_Y \rightarrow \mathbb{R}[s]_1$ takes values in homogeneous linear forms and $a : K_Y \rightarrow \mathbb{R}$ is linear. These coefficient maps depend continuously and semialgebraically on $[Y]$.

Pushforward of moments by $s \mapsto s + t$ transforms a kernel polynomial by $f(s) \mapsto f(s - t)$. Thus translation changes the coefficient maps by

$$b_t(p) = b(p) - \partial_t p, \quad a_t(p) = a(p) - b(p)(t) + p(t), \quad (3.5)$$

where $\partial_t p$ is the directional derivative of p in direction t . Pushforward by $s \mapsto \lambda s$, with $\lambda > 0$, gives

$$b \longmapsto \lambda b, \quad a \longmapsto \lambda^2 a. \quad (3.6)$$

Both actions are congruences on moment matrices. They preserve Hankel structure, rank, and π , and preserve positive semidefiniteness when it is present.

Lemma 3.3. *For every $[Y] \in \mathcal{X}$, the map*

$$\Delta_{K_Y} : \mathbb{R}^7 \longrightarrow \text{Hom}(K_Y, \mathbb{R}[s]_1), \quad t \longmapsto (p \mapsto \partial_t p)$$

is injective, and the coefficient maps a, b in (3.4) are not both zero. The seven infinitesimal translations and the infinitesimal dilation give eight independent variations of the kernel graph. These assertions, including the independence of the variations, also hold over $\mathbf{C}$ under the same rank and multiplication hypotheses.

Proof. If $t \neq 0$ belongs to the kernel of Δ_{K_Y} , a linear change of the pure variables makes t the first coordinate direction. All quadratics in K_Y are then independent of that coordinate, so their products with linear forms cannot span its cube. This contradicts (3.3).

Suppose next that $a = b = 0$, so $J_Y = K_Y$ in the chart. The kernel condition and (3.3) force all moments of degree three to vanish. Multiplying that spanning identity by linear forms shows that $K_Y P = \mathbb{R}[s]_4$, so all moments of degree four vanish as well. With coordinates ordered by degrees 0, 1, 2, the moment matrix consequently has the form

$$\begin{pmatrix} * & * & * \\ * & * & 0 \\ * & 0 & 0 \end{pmatrix},$$

where the degree-zero block has size one and the degree-one block has size seven. The last block row has rank at most one, and the first eight rows have rank at most eight. Thus $\text{rank } Y \leq 9$, a contradiction.

For independence of the infinitesimal variations, suppose a translation in direction t and a dilation with infinitesimal coefficient α have zero combined variation of the graph. Equations (3.5) and (3.6) give

$$\alpha b(p) = \partial_t p, \quad 2\alpha a(p) = b(p)(t) \quad (p \in K_Y).$$

If $\alpha = 0$, injectivity of Δ_{K_Y} gives $t = 0$. If $\alpha \neq 0$, put $c = t/\alpha$. Then $b(p) = \partial_c p$ and $a(p) = p(c)$, so every polynomial in the graph is $p(s + c)$. A translation would therefore produce the already excluded case $a = b = 0$. This proves independence. Each argument is algebraic and applies equally over $\mathbf{C}$. $\square$

Equip the spaces of coefficient maps with the Euclidean inner products induced by the monomial bases. We normalize $[Y]$ in two steps. First choose the unique translation for which $b \perp \text{ran } \Delta_{K_Y}$. Explicitly, its parameter is

$$t = (\Delta_{K_Y}^* \Delta_{K_Y})^{-1} \Delta_{K_Y}^* b.$$

Next choose the unique positive dilation for which

$$\|b\|^2 + \|a\|^2 = 1. \tag{3.7}$$

For the translated coefficients, this amounts to solving $\lambda^2 \|b\|^2 + \lambda^4 \|a\|^2 = 1$. The left side is strictly increasing from zero to infinity, by Lemma 3.3.

Lemma 3.4. *The normalized matrices, together with their data (K_Y, b, a) , form a semialgebraic set $\mathcal{M}$ of dimension 185. Normalization is continuous, and each $[Y] \in \mathcal{X}$ is uniquely recovered from its normalized data and its seven translation parameters and positive dilation parameter. Moreover, $\mathcal{M}$ has a compact ambient semialgebraic space in which the projection to $\text{Gr}(k, P)$ is continuous.*

Proof. The translation formula is continuous because Δ_{K_Y} has constant full column rank. The unique positive solution for λ is continuous and semialgebraic as well. Dilation preserves the orthogonality condition on b , so the two normalizations are compatible. Conversely, start with normalized data, undo its positive dilation, and then undo its translation. Applying the two normalization steps recovers the prescribed parameters. This gives a semialgebraic homeomorphism between $\mathcal{X}$ and $\mathcal{M} \times \mathbb{R}^7 \times \mathbb{R}_{>0}$, and hence $\dim \mathcal{M} = 193 - 8 = 185$.

To specify a compact ambient space, retain the projective matrix $[Y]$ and the subspace K_Y , and extend a, b by zero on $K_Y^\perp$ in P . The extended maps have the same norms as the original maps. Condition (3.7) places their pair on a fixed unit sphere. The product of this sphere with $\mathbf{P}(\mathcal{H})$ and $\text{Gr}(k, P)$ is compact, and projection to the last factor is continuous. We take the closure of $\mathcal{M}$ in this product. $\square$

3.3 Selecting a finite fiber

We now show that the positive semidefinite part of $\mathcal{X}$ has enough distinct projected kernels to avoid both infinite normalized fibers and limits at the boundary of $\mathcal{M}$. We use the standard dimension and fiber-dimension theorems for semialgebraic sets, as well as the fact that the frontier $\overline{S} \setminus S$ of a semialgebraic set has smaller dimension than S ; see [1]. A zero-dimensional semialgebraic set is finite.

At any T representing a point of $\mathcal{X}$, put $J = \ker T$ and $H_0 = \{p : T(p, L) = 0\}$. The kernel of the derivative of the kernel-projection map, before projectivizing, is

$$\{Y' \in \mathcal{H} : Y'(J) \subset T(L)\} = (JH_0)^\perp. \quad (3.8)$$

Here the right side is an annihilator in $\mathbb{R}[x]_4^*$, identified with $\mathcal{H}$. To verify the identity, continue a vector $j \in J$ along a differentiable curve of kernels. Its first variation j' must satisfy

$$Y'j + Tj' = 0.$$

The projected kernel has zero first variation precisely when the pure part of each j' lies in K . Subtracting an element of J then makes j' belong to L , without changing this equation. Thus the condition is $Y'J \subset T(L)$. Conversely, that condition allows all the corrections to be chosen in L ; it also implies $Y'|_{J \times J} = 0$, so Lemma 3.2 gives an actual rank-stratum tangent. Finally, $H_0 = T(L)^\perp$, which proves the annihilator description.

At the seed matrix of Lemma 3.1, the space in (3.8) has dimension 9. The rank stratum before projectivizing has dimension 194, so the derivative of π has rank $194 - 9 = 185$. Projectivization removes one kernel direction, namely scaling, and leaves this derivative rank unchanged. Near the seed matrix, rank remains 20 and its nonzero eigenvalues remain positive. Its L block remains positive definite as well. Consequently the π -image of matrices with these positivity properties has dimension at least 185. It has dimension at most 185 because π factors through $\mathcal{M}$.

The seed's final condition can also be retained in this neighborhood. Indeed, an invertible 321-row product minor for JH_0 remains invertible nearby. On the other hand, scaling, translations, and dilation always give nine independent tangent directions in (3.8): the eight affine directions have independent kernel variations by Lemma 3.3, whereas scaling has zero kernel variation. Thus $\dim(JH_0) \leq 321$ throughout this neighborhood.

Proposition 3.5. *The matrix T in Lemma 3.1 may be chosen so that, for $K = \text{proj}_P \ker T$,*

1. *the fiber $\mathcal{M}_K$ of $\mathcal{M} \rightarrow \text{Gr}(k, P)$ is finite;*
2. *no point of $\overline{\mathcal{M}} \setminus \mathcal{M}$ projects to K .*

Proof. The set of subspaces over which the fiber of $\mathcal{M}$ has positive dimension has dimension at most 184, by the fiber-dimension theorem and $\dim \mathcal{M} = 185$. The frontier of $\mathcal{M}$ has dimension at most 184, so its projection to $\text{Gr}(k, P)$ has dimension at most 184 as well. The positive semidefinite image just constructed has dimension 185. We can therefore choose K in that image outside both exceptional sets, and choose a corresponding positive semidefinite T in the neighborhood under consideration. Its fiber is nonempty and zero-dimensional, hence finite. $\square$

Fix this T and K for the rest of the paper, and set

$$R = K^\perp \subset P, \quad W = L \oplus R. \quad (3.9)$$

Thus $H = W \oplus K$ is an orthogonal decomposition, $\dim R = q = 12$, and $\dim W = d_* = 20$.

Each normalized class $[\bar{Y}] \in \mathcal{M}_K$ determines a nonzero symmetric matrix on W , up to scale. Let $\bar{A}$ be the compression of a representative $\bar{Y}$ to W . The kernel of $\bar{Y}$ is a graph over K with lower terms in L , so W is a complement of that kernel. The same nondegeneracy argument used in Lemma 3.2 shows that $\bar{A}$ is invertible, even if $\bar{Y}$ is indefinite. Define $C_{\bar{Y}} \in \text{Sym}(W)$ to have block form

$$C_{\bar{Y}} = \begin{pmatrix} 0 & 0 \\ 0 & (\bar{A}^{-1})_{R,R} \end{pmatrix} \quad \text{on } W = L \oplus R, \quad \mathcal{C} = \{[C_{\bar{Y}}] : [\bar{Y}] \in \mathcal{M}_K\}. \quad (3.10)$$

If its R, R block were zero, the invertible operator $\bar{A}^{-1}$ would map R into L , which is impossible because $12 > 8$. Thus $\mathcal{C}$ is a finite nonempty set of projective classes of nonzero symmetric matrices on W .

3.4 The inverse-limit property

To connect the normalized fiber to arbitrary limiting sequences, we first need a rank correction that remains small even when some matrix entries diverge. All norms below are fixed Euclidean operator norms; convergence is equivalent in any choice of such norms.

Lemma 3.6 (Uniform rank correction). *Use the fixed decomposition $H = W \oplus K$ from (3.9). Suppose $Y_i \in \mathcal{H}$ have blocks*

$$Y_i = \begin{pmatrix} A_i & B_i \\ B_i^T & D_i \end{pmatrix},$$

where A_i is invertible, $\sup_i \|A_i^{-1}\| < \infty$, and

$$A_i^{-1} B_i \longrightarrow 0, \quad D_i - B_i^T A_i^{-1} B_i \longrightarrow 0.$$

There are Hankel perturbations $\Delta_i \rightarrow 0$ such that $Y_i^\sharp = Y_i + \Delta_i$ has rank d_* . Its W compression $A_i^\sharp$ satisfies

$$(A_i^\sharp)^{-1} - A_i^{-1} \longrightarrow 0,$$

and $\ker Y_i^\sharp \rightarrow K$, or equivalently $\text{ran } Y_i^\sharp \rightarrow W$.

Proof. On matrices whose W block is invertible, write $\mathcal{S}(Y) = D - B^T A^{-1} B$ for the Schur complement on K . For a variation Δ with blocks (a, b, d) and $F = A^{-1} B$,

$$D\mathcal{S}_Y(\Delta) = d - b^T F - F^T b + F^T a F. \quad (3.11)$$

At Y_i , these derivatives tend to the restriction map $\Delta \mapsto \Delta_{K,K}$ on Hankel space. That restriction map is onto $\text{Sym}(K)$ because multiplication on $\text{Sym}^2 K$ is injective. Choose a fixed linear right inverse $\mathcal{E} : \text{Sym}(K) \rightarrow \mathcal{H}$.

We record why an implicit-function argument is uniform in i . Put $M = \sup_i \|A_i^{-1}\|$. On a fixed sufficiently small ball of perturbations, the inverse of the perturbed W block obeys

$$(A_i + a)^{-1} = (I + A_i^{-1}a)^{-1}A_i^{-1}, \quad \|(A_i + a)^{-1}\| \leq 2M.$$

Moreover,

$$(A_i + a)^{-1}(B_i + b) = (I + A_i^{-1}a)^{-1}(A_i^{-1}B_i + A_i^{-1}b)$$

is uniformly bounded there. Differentiating the inverse and this last expression gives

$$\delta(A^{-1}) = -A^{-1}(\delta A)A^{-1}, \quad \delta F = A^{-1}(\delta B - (\delta A)F).$$

Together with (3.11), these identities give a uniform bound for the second derivatives of $\mathcal{S}$ on that ball. This bound uses neither boundedness of A_i nor boundedness of B_i .

Define $f_i(z) = \mathcal{S}(Y_i + \mathcal{E}z)$ for $z \in \text{Sym}(K)$ in a fixed small ball. We have $f_i(0) \rightarrow 0$ and $Df_i(0) \rightarrow I$. Shrinking the fixed ball if needed, the uniform second-derivative bound gives $\|Df_i(z) - I\| \leq 1/2$ there for all sufficiently large i . The map $z \mapsto z - f_i(z)$ is therefore a contraction on the closed ball of radius $2\|f_i(0)\|$: its value at zero has norm $\|f_i(0)\|$, and its Lipschitz constant is at most $1/2$. Its fixed point z_i satisfies $f_i(z_i) = 0$ and $z_i \rightarrow 0$. Set $\Delta_i = \mathcal{E}z_i$.

The W block remains invertible, and its Schur complement is zero, so $\text{rank } Y_i^\sharp = d_*$. The inverse identity gives $(A_i^\sharp)^{-1} - A_i^{-1} \rightarrow 0$. The displayed formula for the perturbed cross block also gives $(A_i^\sharp)^{-1}B_i^\sharp \rightarrow 0$. The kernel of $Y_i^\sharp$ is the graph with columns $(\begin{smallmatrix} -(A_i^\sharp)^{-1}B_i^\sharp \\ I_K \end{smallmatrix})$, proving the final assertion. Discarding finitely many indices has no effect on any of the conclusions. $\square$

Lemma 3.7 (Finite inverse limits). *Let T, K, R, W , and $\mathcal{C}$ be chosen as above. Suppose a sequence $Y_i \in \mathcal{H}$ has a group of d_* eigenvalues bounded away from zero in absolute value, and the span of the corresponding eigenvectors tends to W . Suppose all remaining eigenvalues tend to zero. If the inverses of the compressions $A_i = (Y_i)_{W,W}$ converge to a nonzero matrix C , then*

$$[C] \in \mathcal{C}.$$

Proof. We first put the sequence in the situation of Lemma 3.6. Split off the distinguished eigenvalues to write $Y_i = Y_i^{\text{large}} + E_i$, where $\text{rank } Y_i^{\text{large}} = d_*$ and $E_i \rightarrow 0$. After passage to a subsequence, orthonormal frames for the large eigenspaces have W components S_i converging to an orthogonal matrix and K components tending to zero. In particular S_i is invertible, and

$$Y_i^{\text{large}} = \begin{pmatrix} I \\ G_i \end{pmatrix} A_i^0 \begin{pmatrix} I & G_i^T \end{pmatrix}, \quad G_i \rightarrow 0, \quad \sup_i \|(A_i^0)^{-1}\| < \infty. \quad (3.12)$$

Indeed $A_i^0 = S_i \text{diag}(\lambda_{i,1}, \dots, \lambda_{i,d_*}) S_i^T$, and the inverse eigenvalues are bounded.

Write the blocks of E_i as (a_i, b_i, d_i) in the same order as in the rank-correction lemma. Then

$$A_i = A_i^0 + a_i, \quad B_i = A_i^0 G_i^T + b_i, \quad D_i = G_i A_i^0 G_i^T + d_i.$$

The inverses A_i^{-1} are bounded, and

$$A_i^{-1}B_i = G_i^T + A_i^{-1}(b_i - a_i G_i^T) \rightarrow 0.$$

With $e_i = b_i - a_i G_i^T$, direct block elimination gives

$$D_i - B_i^T A_i^{-1} B_i = d_i - G_i b_i - b_i^T G_i^T + G_i a_i G_i^T - e_i^T A_i^{-1} e_i \rightarrow 0.$$

These formulas explain explicitly why divergence of the large eigenvalues causes no difficulty. Apply Lemma 3.6. The corrected Hankel matrices have exact rank d_* , their kernels tend to K , and their compression inverses have the same nonzero limit C . We replace Y_i by these corrected matrices for the rest of the proof.

For large i , the class $[Y_i]$ belongs to $\mathcal{X}$: its kernel is close to $K \subset H$, and the required product conditions are open at K . Put $K_i = \pi([Y_i])$. Then $K_i \rightarrow K$, and the lower coefficient maps $a_i^{\ker}, b_i^{\ker}$ of the kernel graph tend to zero. The translations used in normalization consequently tend to zero, by the explicit left-inverse formula for Δ_{K_i} . The translated coefficient maps still tend to zero, so the normalizing dilation factors tend to infinity. Let their reciprocals be $\varepsilon_i \rightarrow 0$.

The normalized data lie in the compact ambient space of Lemma 3.4. Pass to a convergent subsequence. Its limit projects to K , and Proposition 3.5 excludes a limit in the frontier of $\mathcal{M}$. It is therefore a member $[\bar{Y}] \in \mathcal{M}_K$. Choose representatives Z_i of the normalized matrices converging to a representative $\bar{Y}$. Up to nonzero scalar multiples, the original matrices are obtained from Z_i by dilation by ε_i , followed by translations whose parameters tend to zero.

We compute the inverse limit first before those translations. Set $R_i = K_i^\perp \subset P$ and $W_i = L \oplus R_i$. Use orthonormal coordinates on W_i converging to the fixed coordinates on W , and keep the coordinates on L unchanged. The compressions $\bar{A}_i = (Z_i)_{W_i, W_i}$ then converge to the invertible compression $\bar{A}$ of $\bar{Y}$ to W . Dilation acts on monomial columns by

$$D_\varepsilon = \text{diag}(1, \varepsilon I_7, \varepsilon^2 I_{28})$$

on H , according to degrees 0, 1, 2. Since it preserves W_i , the compression of the dilated matrix is

$$A_i^{\text{dil}} = D_{\varepsilon_i, W_i} \bar{A}_i D_{\varepsilon_i, W_i}.$$

Consequently

$$\varepsilon_i^4 (A_i^{\text{dil}})^{-1} = \text{diag}(\varepsilon_i^2, \varepsilon_i I_7, I_q) \bar{A}_i^{-1} \text{diag}(\varepsilon_i^2, \varepsilon_i I_7, I_q) \longrightarrow C_{\bar{Y}}. \quad (3.13)$$

It remains to justify changing W_i to W and restoring the translations. After dilation, the kernel coefficients of Z_i are $\varepsilon_i b_i^{\text{norm}}$ and $\varepsilon_i^2 a_i^{\text{norm}}$. The normalized coefficients are bounded. Write the dilated kernel in $W_i \oplus K_i$ as $\{(F_i k, k) : k \in K_i\}$; the preceding coefficient bounds give $F_i \rightarrow 0$. Its orthogonal complement, which is the range of the dilated matrix, has graph embedding

$$\iota_i : W_i \longrightarrow H, \quad \iota_i w = (w, -F_i^* w).$$

Thus ι_i tends to the inclusion of W after the chosen coordinate identifications, and the full dilated matrix is $\iota_i A_i^{\text{dil}} \iota_i^T$.

Let B_i denote the monomial matrix of the remaining translation. Its parameter tends to zero, so $B_i \rightarrow I_H$. Compression to W after this translation therefore gives

$$M_i A_i^{\text{dil}} M_i^T, \quad M_i = \text{proj}_W B_i \iota_i \longrightarrow I_W.$$

The actual compression A_i differs from this matrix by a nonzero scalar α_i , coming from the choice of projective representative. Equation (3.13) yields

$$\alpha_i \varepsilon_i^4 A_i^{-1} = M_i^{-T} (\varepsilon_i^4 (A_i^{\text{dil}})^{-1}) M_i^{-1} \longrightarrow C_{\bar{Y}} \neq 0.$$

Hence $[A_i^{-1}] \rightarrow [C_{\bar{Y}}]$. Since $A_i^{-1} \rightarrow C \neq 0$ also gives $[A_i^{-1}] \rightarrow [C]$, we obtain $[C] = [C_{\bar{Y}}] \in \mathcal{C}$. $\square$

We retain the positive semidefinite matrix T , the subspaces $K, R = K^\perp \subset P$, and $W = L \oplus R$, and the finite set $\mathcal{C}$ from (3.10). These are the geometric data used in the next construction.

4 Separation from Hankel images

Retain the Hankel matrix T , the spaces K , $R = K^\perp \subset P$, and $W = L \oplus R$ from the preceding section. In particular, $\dim W = d_* = 20$ and $H = W \oplus K$ orthogonally. Let $\mathcal{C}$ be the finite set of projective classes of nonzero symmetric operators on W supplied by Lemma 3.7. We will construct a vector w and a subspace U such that w is outside the closure of all images of U obtained by applying one Hankel matrix componentwise. A quadratic form separating w from those images will then give the required positive semidefinite quadratic matrix Q .

Fix an integer $N > 10h^2$, where $h = \dim H = 36$, and put

$$r = d_*N - 1. \quad (4.1)$$

We equip H^N with the sum of the Euclidean inner products on its factors. For a subspace $F \subset H$, write P_F for orthogonal projection onto F ; the notation P_F^N denotes its componentwise action on H^N . The two subspaces in the third condition below will represent directions in which eigenvalues diverge or have finite nonzero limits. Their different projection constraints arise by taking limits of bounded Hankel images of unit tuples.

Lemma 4.1 (Choice of the vector and subspace). *There exist a unit vector $w = (w_1, \dots, w_N) \in W^N$ and an r -dimensional subspace $U \subset w^\perp \subset H^N$ with the following properties.*

1. *The vectors $w_1, \dots, w_N$ span W , and*

$$\sum_{i=1}^N w_i^T C w_i \neq 0 \quad \text{whenever } [C] \in \mathcal{C}. \quad (4.2)$$

2. *The map*

$$P_W^N|_U : U \longrightarrow w^\perp \cap W^N \quad (4.3)$$

is an isomorphism.

3. *Suppose that $F_\infty, F_0 \subset H$ are orthogonal subspaces, C_0 is a symmetric operator on F_0 , and $\dim(F_\infty \oplus F_0) > d_*$. No nonzero tuple $u = (u_1, \dots, u_N) \in U$ satisfies*

$$P_{F_\infty} u_i = 0, \quad P_{F_0} u_i = C_0 P_{F_0} w_i \quad (1 \leq i \leq N). \quad (4.4)$$

Proof. For a nonzero symmetric operator C on W , the polynomial $\sum_i w_i^T C w_i$ on W^N is nonzero: a vector with just one nonzero component already detects a nonzero value of the quadratic form of C . Its nonvanishing therefore defines a dense open subset of W^N . The spanning condition is also dense and open, since $N \geq \dim W$. There are only finitely many classes in $\mathcal{C}$, and changing a representative does not affect nonvanishing. We may consequently choose w satisfying all these conditions and then normalize it to unit length.

Fix this w and consider the Grassmannian

$$\mathcal{G} = \text{Gr}(r, w^\perp), \quad \dim w^\perp = hN - 1.$$

Inside $w^\perp$ we have the orthogonal decomposition

$$w^\perp = (w^\perp \cap W^N) \oplus K^N.$$

The r -planes complementary to K^N form a nonempty open subset of $\mathcal{G}$. They are exactly the subspaces satisfying (4.3).

We show that the subspaces violating the third property have smaller dimension than $\mathcal{G}$. Fix dimensions for F_∞ and F_0 , and set $j = \dim F_\infty + \dim F_0 > d_*$. The triples (F_∞, F_0, C_0) form a semialgebraic parameter space of dimension at most $4h^2$. For example, orthogonal projections onto the two spaces and the extension of C_0 by zero give matrix coordinates with this bound. For each fixed triple, (4.4) prescribes the component of every u_i in $F_\infty \oplus F_0$. Its solution set in H^N is therefore an affine space of dimension $(h - j)N$.

For a fixed nonzero $u \in w^\perp$, the subspaces in $\mathcal{G}$ containing u form a Grassmannian of codimension

$$(hN - 1) - r = (h - d_*)N$$

in $\mathcal{G}$. Apply the semialgebraic dimension bounds to the incidence relation consisting of a parameter triple, a nonzero solution $u \in w^\perp$ of (4.4), and an r -plane containing u . Its image in $\mathcal{G}$ has dimension at most

$$\begin{aligned} \dim \mathcal{G} + 4h^2 + (h - j)N - (h - d_*)N &= \dim \mathcal{G} + 4h^2 - (j - d_*)N \\ &< \dim \mathcal{G}. \end{aligned}$$

There are finitely many possible pairs of dimensions for F_∞, F_0 . The union of their exceptional images still has smaller dimension and cannot contain the nonempty open set of complements to K^N . Choosing U outside that union proves both remaining properties. $\square$

Fix w and U as in Lemma 4.1. Notice that the P^N component of w is nonzero: otherwise all its components would lie in L and could not span W . We now use the third property of the lemma to control every possible limit of Hankel images, including limits produced by unbounded matrices.

Lemma 4.2 (Separation from the closure). *For w and U as in Lemma 4.1, let*

$$\mathcal{S}_U = \{(I_N \otimes Y)u : u \in U, Y \text{ a Hankel matrix on } H\}.$$

Then $w \notin \overline{\mathcal{S}_U}$.

Proof. Suppose, to the contrary, that there are tuples $u^{(n)} \in U$ and Hankel matrices Y_n such that

$$(I_N \otimes Y_n)u^{(n)} \longrightarrow w.$$

After omitting finitely many terms the tuples are nonzero. Replacing $u^{(n)}$ by $u^{(n)}/\|u^{(n)}\|$ and Y_n by $\|u^{(n)}\|Y_n$ leaves the products unchanged. We may thus assume that $\|u^{(n)}\| = 1$. Pass to a subsequence for which $u^{(n)} \rightarrow u \in U$, where $\|u\| = 1$.

Choose an orthonormal eigenbasis $e_{1,n}, \dots, e_{h,n}$ for each Y_n , with eigenvalues $\lambda_{1,n}, \dots, \lambda_{h,n}$. After a further subsequence, each eigenvector converges and each eigenvalue converges in the extended real line. The limiting eigenvectors form an orthonormal basis of H . Let F_∞ be the span of those corresponding to eigenvalues whose absolute values tend to infinity, and let F_0 be the span of those corresponding to finite nonzero limits. All remaining eigenvalues tend to zero. Define C_0 on F_0 by taking the reciprocal of each corresponding eigenvalue limit in this limiting basis.

For each component of the tuples, the spectral identity is

$$\langle e_{\ell,n}, Y_n u_i^{(n)} \rangle = \lambda_{\ell,n} \langle e_{\ell,n}, u_i^{(n)} \rangle. \quad (4.5)$$

When $\lambda_{\ell,n} \rightarrow 0$, the right side tends to zero because $u^{(n)}$ is bounded. It follows that each w_i belongs to $F := F_\infty \oplus F_0$, and the spanning property of w gives $W \subset F$. When $|\lambda_{\ell,n}| \rightarrow \infty$, divide (4.5)

by $\lambda_{\ell,n}$ to see that the corresponding component of u_i is zero. For finite nonzero limits, the same identity gives

$$P_{F_\infty} u_i = 0, \quad P_{F_0} u_i = C_0 P_{F_0} w_i \quad (1 \leq i \leq N).$$

Since $u \neq 0$, Lemma 4.1 excludes $\dim F > d_*$. Together with $W \subset F$ this yields $F = W$. Moreover F_0 is nonzero. Otherwise $F_\infty = W$ and $P_W^N u = 0$, contradicting the injectivity in (4.3).

Extend C_0 by zero on F_∞ to obtain a nonzero symmetric operator C on W . We next verify that C is an inverse limit covered by Lemma 3.7. The two groups defining F consist of exactly d_* eigenvalues, bounded away from zero in absolute value, and the span of their eigenvectors converges to W . All remaining eigenvalues tend to zero. Let

$$A_n = P_W Y_n|_W$$

be the compression to W . To identify its inverse, first keep only the d_* eigenvalues in the two groups. In a fixed orthonormal basis of W , their contribution to A_n is

$$A_n^{\text{large}} = S_n \Lambda_n S_n^T,$$

where the columns of S_n are the projected eigenvectors and Λ_n is the diagonal matrix of these eigenvalues. The matrices S_n converge to an orthogonal matrix, while Λ_n^{-1} converges to the diagonal matrix of the reciprocal finite limits and zeros for the divergent eigenvalues. Consequently

$$(A_n^{\text{large}})^{-1} \longrightarrow C.$$

The contribution of the remaining eigenvalues tends to zero in norm. Since the displayed inverses are bounded, the identity

$$A_n^{-1} = (I + (A_n^{\text{large}})^{-1}(A_n - A_n^{\text{large}}))^{-1} (A_n^{\text{large}})^{-1}$$

shows that A_n is invertible for all sufficiently large n and $A_n^{-1} \rightarrow C$. Lemma 3.7 now gives $[C] \in \mathcal{C}$.

The projection identities above say $P_W u_i = C w_i$. Since $u \in U$ and $U \subset w^\perp$, we obtain

$$0 = \langle w, u \rangle = \sum_{i=1}^N \langle w_i, P_W u_i \rangle = \sum_{i=1}^N w_i^T C w_i.$$

This contradicts (4.2) and proves the lemma. $\square$

The preceding separation has a useful quadratic consequence. The set $\mathcal{S}_U$ is closed under multiplication by any real scalar. Hence its closed set of directions on the unit sphere contains neither w nor $-w$. Compactness gives a number $\delta > 0$ such that

$$\|P_{w^\perp} z\| \geq \delta \|z\| \quad (z \in \mathcal{S}_U).$$

Here the projections act on H^N , and w has unit length. Define the symmetric operator

$$g = \delta^{-2} P_{w^\perp} - P_{\mathbb{R}w}.$$

For $z \in \mathcal{S}_U$, its quadratic form satisfies $z^T g z \geq \|z\|^2 - \|P_{\mathbb{R}w} z\|^2 \geq 0$, whereas $w^T g w = -1$. We have therefore obtained

$$w^T g w < 0, \quad ((I_N \otimes Y)u)^T g ((I_N \otimes Y)u) \geq 0 \quad (u \in U, Y \text{ Hankel}). \quad (4.6)$$

We can now define the quadratic matrix used in the rest of the proof. Use the monomial coordinates to identify $\mathbb{R}[x_0, \dots, x_7]_4^*$ with $\mathbb{R}^m$, where $m = 330$. For such a functional y , let $Y(y)$ be its Hankel matrix on H , so $Y(y)(p, p') = y(pp')$, and set

$$G(y) = I_N \otimes Y(y).$$

Choose a basis of U and also write U for the hN -by- r matrix whose columns are these basis vectors. Define

$$Q(y) = U^T G(y) g G(y) U. \quad (4.7)$$

The entries of $G(y)$ are linear in y , so Q is a real symmetric r -by- r matrix with homogeneous quadratic entries. For every $a \in \mathbb{R}^r$, the vector Ua belongs to the subspace U , and (4.6) gives

$$a^T Q(y) a = (G(y) U a)^T g (G(y) U a) \geq 0.$$

Thus $Q(y) \succeq 0$ for every real y . The data retained for the next section are T , w , U , and g , together with this admissible map Q ; the strict inequality at w in (4.6) will produce the obstruction to a finite semidefinite lift.

5 A local obstruction to a semidefinite lift

We prove that the quadratic matrix map constructed in Section 4 has an epigraph set E_Q with no finite semidefinite lift. The proof has two parts. First, a polynomial patch in E_Q admits a positive moment functional whose application at small scales violates the defining inequality of E_Q to order four. Second, any finite lift would force the resulting points to lie much closer to E_Q than this violation allows. The second part is a local sum-of-squares obstruction, in the sense of the methods developed by Scheiderer [18, Section 4]; here we give a direct proof using finite Taylor polynomials of Gram factors.

We retain the spaces $H = L \oplus P$, $R = K^\perp \subset P$, and $W = L \oplus R$, the positive semidefinite Hankel matrix T , and the data N, w, U, g of the preceding sections. In particular,

$$\dim H = 36, \quad \dim L = 8, \quad \dim R = 12, \quad \text{rank } T = 20, \quad r = 20N - 1.$$

Write $J = \ker T$. We use that $J \cap L = 0$ and that the projection of J to P is K . As in Section 4, the symbol U also denotes a $36N$ -by- r matrix whose columns form a basis of the subspace $U \subset H^N$. By Lemma 4.1, orthogonal projection induces an isomorphism

$$U \longrightarrow w^\perp \cap W^N. \quad (5.1)$$

Moreover, $w \in W^N$ has components spanning W , and $w^T g w < 0$. The admissible quadratic matrix map is $Q(y) = U^T G(y) g G(y) U$, as defined in (4.7).

To keep track of repeated blocks, for an operator S on H we write $S_N = I_N \otimes S$, an operator on H^N . The same convention applies to maps between subspaces of H . We order the coordinates of H^N as $L^N \oplus P^N$ when using two-by-two block matrices. Subscripts LL , LP , and PP on an operator on H^N refer to these blocks.

5.1 A polynomial patch and its moment deformation

Let $v(s) \in H$, for $s \in \mathbb{R}^7$, be the column of homogeneous quadratic monomials evaluated at $(1, s)$. Thus its entries have affine degrees zero, one, and two. Let y_s be evaluation at $(1, s)$ on homogeneous quartics, and define

$$V(s) = I_N \otimes v(s) \in \mathbb{R}^{36N \times N}, \quad M(s) = U^T V(s) \in \mathbb{R}^{r \times N}. \quad (5.2)$$

The moment identity gives $G(y_s) = V(s)V(s)^T$. In the following lemma, E_Q is the set defined in Proposition 2.2.

Lemma 5.1 (Polynomial patch). *There is an open ball $\Omega_0 \subset \mathbb{R}^7$ about zero such that $M(s)$ has rank N for every $s \in \Omega_0$. The polynomial map*

$$x(s) = (M(s)M(s)^T, \operatorname{tr}(gG(y_s)), y_s) \quad (5.3)$$

has degree at most four and takes Ω_0 into E_Q . Its values satisfy the trace inequality defining E_Q with equality.

Proof. Decompose $w = w_L + w_P$ in $L^N \oplus P^N$. Since the components of w span W and $R \neq 0$, we have $w_P \neq 0$; also $w_P \in R^N$. Given $\ell \in L^N$, the vector

$$\ell - \frac{\langle \ell, w_L \rangle}{\|w_P\|^2} w_P$$

belongs to $w^\perp \cap W^N$. By (5.1), it is the projection to W^N of some vector of U . Thus projection from U to L^N is surjective. The columns of $V(0)$ form a basis for the N constant-coordinate directions in L^N . If $M(0)a = 0$, then $V(0)a$ is orthogonal to U ; surjectivity onto L^N forces $V(0)a = 0$, and hence $a = 0$. Therefore $\operatorname{rank} M(0) = N$, and the same holds on a sufficiently small open ball.

Put $\Theta(s) = V(s)^T g V(s)$. The definition of Q gives

$$Q(y_s) = M(s)\Theta(s)M(s)^T.$$

Consequently

$$\operatorname{ran} Q(y_s) \subset \operatorname{ran} M(s) = \operatorname{ran}(M(s)M(s)^T).$$

Full column rank of $M(s)$ gives

$$M(s)^T (M(s)M(s)^T)^\dagger M(s) = I_N,$$

and hence

$$\operatorname{tr}((M(s)M(s)^T)^\dagger Q(y_s)) = \operatorname{tr} \Theta(s) = \operatorname{tr}(gG(y_s)).$$

This proves membership in E_Q . Finally, $M(s)$ has degree at most two, while $G(y_s)$ and y_s have degree at most four, proving the degree bound. $\square$

For a linear functional Λ on polynomials in seven variables through degree four, its moment matrix is the symmetric matrix indexed by monomials of degree at most two, with (α, β) entry $\Lambda(s^{\alpha+\beta})$. Under the affine chart $x_0 = 1$, this is precisely a Hankel matrix on H . Applying Λ to a vector or matrix of polynomials always means applying it entrywise.

Lemma 5.2 (A deformation outside E_Q). *There exist a linear functional $\Lambda : \mathbb{R}[s_1, \dots, s_7]_{\leq 4} \rightarrow \mathbb{R}$ and an open ball $\Omega \subset \Omega_0$ about zero with the following properties:*

1. $\Lambda(1) = 1$, and its moment matrix on polynomials of degree at most two is positive definite;
2. for every $c \in \Omega$, writing

$$(A_{c,\varepsilon}, t_{c,\varepsilon}, y_{c,\varepsilon}) = \Lambda_s(x(c + \varepsilon s)), \quad (5.4)$$

there are constants $a_c > 0$ and $\kappa_c > 0$ such that, as $\varepsilon \downarrow 0$,

$$A_{c,\varepsilon} \succeq a_c \varepsilon^4 I_r, \quad (5.5)$$

$$t_{c,\varepsilon} - \operatorname{tr}(A_{c,\varepsilon}^{-1} Q(y_{c,\varepsilon})) = -\kappa_c \varepsilon^4 + o(\varepsilon^4). \quad (5.6)$$

All three entries of (5.4) remain bounded as $\varepsilon \downarrow 0$.

Proof. We first allow the moment matrix to be an arbitrary Hankel matrix S near T , and normalize its constant moment at the end. Let B_c be the matrix on H determined by $B_cv(s) = v(s+c)$. This is an invertible matrix, depends polynomially on c , and satisfies $B_0 = I_H$. In the affine chart, L consists of the constant and linear monomials, and P consists of the quadratic monomials. On this degree decomposition of H , put

$$D_\varepsilon = \text{diag}(1, \varepsilon I_7, \varepsilon^2 I_{28}).$$

The functional with moment matrix S , applied at $c + \varepsilon s$, has moment matrix $B_c D_\varepsilon S D_\varepsilon B_c^T$. Its repeated matrix is therefore

$$G = B_{c,N} D_{\varepsilon,N} S_N D_{\varepsilon,N} B_{c,N}^T. \quad (5.7)$$

For the corresponding tested data, $A = U^T G U$ and $t = \text{tr}(gG)$. Whenever A is invertible, the quantity to be made negative is

$$t - \text{tr}(A^{-1}Q(y)) = \text{tr}(g[G - GU(U^T GU)^{-1}U^T G]). \quad (5.8)$$

We now analyze the matrix inside this trace at small ε . Projection of the column space of $B_{c,N}^T U$ to L^N remains surjective for c near zero. We can therefore choose a continuously varying basis matrix U_c for this column space in the form

$$U_c = \begin{pmatrix} I_{8N} & 0 \\ F_c & E_c \end{pmatrix}, \quad F_c \in \mathbb{R}^{28N \times 8N}, \quad E_c \in \mathbb{R}^{28N \times (12N-1)}. \quad (5.9)$$

For completeness, keep one invertible $8N$ -by- $8N$ minor of the projection matrix and normalize its columns to have L^N components I_{8N} . From each remaining column subtract its L^N component expressed in these normalized columns. The resulting columns have zero L^N component and give the block E_c . All these operations depend continuously on c . At $c = 0$, projection of the columns of E_0 to R^N is injective: a vector $(0, E_0 z)$ with zero R^N projection would belong to $U \cap K^N$, which is zero by (5.1).

Let $D_{\varepsilon,L,N}$ denote the restriction of $D_{\varepsilon,N}$ to L^N . For $\varepsilon > 0$, multiplication of $D_{\varepsilon,N} U_c$ on the right by the invertible block diagonal matrix with blocks $D_{\varepsilon,L,N}^{-1}$ and $\varepsilon^{-2} I_{12N-1}$ gives

$$\hat{U}_{c,\varepsilon} = \begin{pmatrix} I_{8N} & 0 \\ \varepsilon^2 F_c D_{\varepsilon,L,N}^{-1} & E_c \end{pmatrix}. \quad (5.10)$$

This matrix extends continuously to $\varepsilon = 0$, where the lower-left block is zero. The Gram matrix $\hat{U}_{0,0}^T T_N \hat{U}_{0,0}$ is positive definite. Indeed, if a vector $(\ell, E_0 z)$ belongs to J^N , then $E_0 z \in K^N$. Injectivity of its R^N projection gives $z = 0$, and $J \cap L = 0$ then gives $\ell = 0$. Thus the column space of $\hat{U}_{0,0}$ meets $\ker T_N$ trivially.

It follows that, for (c, ε, S) near $(0, 0, T)$, the Gram matrix in

$$\hat{C}_{c,\varepsilon}(S) = S_N - S_N \hat{U}_{c,\varepsilon} (\hat{U}_{c,\varepsilon}^T S_N \hat{U}_{c,\varepsilon})^{-1} \hat{U}_{c,\varepsilon}^T S_N \quad (5.11)$$

is invertible. In particular, the displayed residual is continuous in all three arguments, including at $\varepsilon = 0$. Changing a basis of the columns does not change the residual in (5.8). After removing the outer translation factors, define

$$C_{c,\varepsilon}(S) = D_{\varepsilon,N} \hat{C}_{c,\varepsilon}(S) D_{\varepsilon,N}.$$

The exact relation to the matrix in (5.8) is

$$G - GU(U^T GU)^{-1}U^T G = B_{c,N} C_{c,\varepsilon}(S) B_{c,N}^T. \quad (5.12)$$

The matrix $C_{c,\varepsilon}(S)$ annihilates U_c . If a symmetric block matrix C annihilates the first $8N$ columns $\begin{pmatrix} I_{8N} \\ F_c \end{pmatrix}$ of U_c , its blocks satisfy $C_{PL} = -C_{PP}F_c$ and $C_{LL} = F_c^T C_{PP} F_c$. Hence, with

$$\iota_c = \begin{pmatrix} -F_c^T \\ I_{28N} \end{pmatrix} : P^N \longrightarrow H^N,$$

we obtain the exact identity

$$C_{c,\varepsilon}(S) = \varepsilon^4 \iota_c(\widehat{C}_{c,\varepsilon}(S))_{PP} \iota_c^T \quad (\varepsilon > 0). \quad (5.13)$$

Combining (5.12) and (5.13), the limit of (5.8) divided by ε^4 is

$$\text{tr}(g B_{c,N} \iota_c(\widehat{C}_{c,0}(S))_{PP} \iota_c^T B_{c,N}^T). \quad (5.14)$$

This expression is continuous in c and S . Its sign can therefore be computed first at the singular moment matrix T .

At $(c, \varepsilon, S) = (0, 0, T)$, the matrix in (5.11) is positive semidefinite of rank $20N - r = 1$. To see this rank directly, set $Z = T_N^{1/2} \widehat{U}_{0,0}$. The residual equals $T_N^{1/2}(I - P_Z)T_N^{1/2}$, where P_Z is the orthogonal projection onto the r -dimensional column space of Z , a subspace of $\text{ran } T_N^{1/2}$. It annihilates L^N , the embedded columns $\begin{pmatrix} 0 \\ E_0 \end{pmatrix}$, and J^N . Its range is consequently contained in

$$P^N \cap (J^N)^\perp \cap (\text{ran } E_0)^\perp = R^N \cap (\text{ran } E_0)^\perp.$$

Because projection of E_0 to R^N has rank $12N - 1$, this last space is one-dimensional. It contains w_P , since w is orthogonal to every column of U_0 . Therefore

$$(\widehat{C}_{0,0}(T))_{PP} = \alpha w_P w_P^T \quad \text{for some } \alpha > 0.$$

The other columns of U_0 give $w_L = -F_0^T w_P$, so $\iota_0 w_P = w$. Restoring the translation in (5.13), whose matrix is the identity at $c = 0$, shows that the limiting coefficient in (5.8) is

$$\alpha \text{tr}(g w w^T) = \alpha w^T g w < 0.$$

Choose a positive definite Hankel matrix S sufficiently close to T . Such a choice is possible by adding a small positive multiple of the degree-two moment matrix of a nondegenerate Gaussian measure on $\mathbb{R}^7$. By the continuity just proved, the limiting coefficient remains negative for every c in an open ball about zero. Its constant moment is positive. Normalize the corresponding functional by its value at 1 and call the result Λ . Scaling the functional by a positive number ρ scales A, t, y by ρ , and scales $Q(y)$ by ρ^2 . It therefore scales (5.8) by ρ , preserving the sign. Shrinking the ball to lie in Ω_0 proves (5.6).

Let $S_\Lambda > 0$ be the normalized moment matrix. For fixed c , the matrix $B_{c,N}^T U$ has full column rank. Since the smallest diagonal entry of $D_{\varepsilon,N}$ is ε^2 for $0 < \varepsilon \leq 1$, formula (5.7) gives, for $z \in \mathbb{R}^r$,

$$z^T A_{c,\varepsilon} z \geq \lambda_{\min}(S_\Lambda) \varepsilon^4 \|B_{c,N}^T U z\|^2 \geq a_c \varepsilon^4 \|z\|^2$$

with $a_c > 0$. Finally, the tested coordinates are polynomials in ε , so they remain bounded. This completes the proof. $\square$

The preceding lemma supplies points outside E_Q whose violation has a controlled leading term. We next show that a semidefinite lift would force any such moment deformation of a polynomial patch to be arbitrarily well approximated, to a prescribed finite order, by feasible points.

5.2 Moment tests of analytic lift data

We first record why moments given only through degree four can be used to test Taylor polynomials of higher degree. The assertion concerns finite positive moment matrices and does not require a representing measure.

Lemma 5.3 (Finite positive extension). *Let $n \geq 1$ and $j \geq 0$ be integers, let $z = (z_1, \dots, z_n)$, and let the linear functional $\lambda : \mathbb{R}[z]_{\leq 2j} \rightarrow \mathbb{R}$ satisfy $\lambda(p^2) > 0$ for every nonzero $p \in \mathbb{R}[z]_{\leq j}$. For every integer $b \geq j$, there is an extension $\tilde{\lambda} : \mathbb{R}[z]_{\leq 2b} \rightarrow \mathbb{R}$ such that $\tilde{\lambda}(p^2) > 0$ for every nonzero $p \in \mathbb{R}[z]_{\leq b}$.*

Proof. It suffices to extend from j to $j + 1$. Index the enlarged moment matrix first by monomials of degree at most j , and then by monomials homogeneous of degree $j + 1$. It has the form

$$\begin{pmatrix} M_j & C \\ C^T & D \end{pmatrix}, \quad M_j \succ 0.$$

The mixed block C uses moments of degree at most $2j + 1$. Its entries of degree at most $2j$ are already fixed, and we may assign arbitrary values to all moments of degree $2j + 1$. Every entry of D has degree exactly $2j + 2$, so none of these moments has yet been assigned. Give them the values $a \int z^\gamma d\gamma_n(z)$, where γ_n is a nondegenerate Gaussian probability measure and $a > 0$. Then $D = aD_0$, where $D_0 \succ 0$: the integral of the square of any nonzero homogeneous polynomial is strictly positive. Choosing a sufficiently large makes $aD_0 - C^T M_j^{-1} C$ positive definite. The Schur complement proves positivity of the enlarged moment matrix. Each degree layer is assigned as a functional on monomials, so all repeated Hankel entries agree. Iteration proves the assertion. $\square$

The next lemma provides simultaneous analytic lift variables and Gram factors. Only a nonempty open part of the polynomial patch is needed.

Lemma 5.4 (Local analytic Gram factors). *Let $C \subset \mathbb{R}^a$ have an exact representation*

$$C = \{x \in \mathbb{R}^a : \text{there exists } u \in \mathbb{R}^\nu \text{ with } \mathcal{L}(x, u) \succeq 0\},$$

where $\nu \geq 0$ and $\mathcal{L}$ is an affine symmetric pencil of size $\ell \geq 1$. Let $\Omega \subset \mathbb{R}^n$ be nonempty, open, and semialgebraic, and let $p : \mathbb{R}^n \rightarrow \mathbb{R}^a$ be polynomial with $p(\Omega) \subset C$. There is a nonempty open subset $\Omega' \subset \Omega$ and real analytic maps $u : \Omega' \rightarrow \mathbb{R}^\nu$ and $F : \Omega' \rightarrow \mathbb{R}^{\ell \times \ell}$ such that

$$\mathcal{L}(p(s), u(s)) = F(s)F(s)^T \quad (s \in \Omega'). \quad (5.15)$$

Proof. Consider the semialgebraic set

$$\mathcal{A} = \{(s, u, F) : s \in \Omega, \quad \mathcal{L}(p(s), u) = FF^T\}.$$

Its projection to Ω is surjective, since every positive semidefinite matrix has a real square Gram factor. A semialgebraic set admits a finite stratification by real analytic semialgebraic manifolds; see [1, Proposition 9.1.8], or [14, Section 2.2.2]. On at least one stratum, the differential of the projection has rank n at some point. Otherwise Sard's theorem on each of the finitely many strata would imply that the whole image has Lebesgue measure zero, contradicting that it contains Ω . The analytic submersion theorem at such a point gives a local analytic section of the projection. Its u and F coordinates give (5.15). $\square$

Lemma 5.5 (Feasible approximation after a moment test). *Let C , p , and Ω be as in Lemma 5.4. Fix an integer $b \geq \max\{1, \deg p\}$ and a linear functional $\lambda : \mathbb{R}[z_1, \dots, z_n]_{\leq 2b} \rightarrow \mathbb{R}$ satisfying $\lambda(1) = 1$ and $\lambda(q^2) \geq 0$ for all $q \in \mathbb{R}[z]_{\leq b}$. There is a nonempty open subset $\Omega' \subset \Omega$ such that, for each $c \in \Omega'$, the points*

$$x_\varepsilon = \lambda_z(p(c + \varepsilon z))$$

admit points $\tilde{x}_\varepsilon \in C$, for all sufficiently small $\varepsilon > 0$, with

$$\|\tilde{x}_\varepsilon - x_\varepsilon\| = O(\varepsilon^b). \quad (5.16)$$

The implied constant may depend on c and on the displayed data.

Proof. Choose analytic maps u, F on Ω' as in Lemma 5.4, and fix $c \in \Omega'$. For a real analytic map f , write its degree- b Taylor polynomial at c , evaluated at $c + \varepsilon z$, as

$$f_{c,b}(\varepsilon, z) = \sum_{|\alpha| \leq b} \frac{\partial^\alpha f(c)}{\alpha!} \varepsilon^{|\alpha|} z^\alpha.$$

Set $u_\varepsilon = \lambda_z(u_{c,b}(\varepsilon, z))$ and

$$P_\varepsilon = \mathcal{L}(x_\varepsilon, u_\varepsilon), \quad H_\varepsilon = \lambda_z(F_{c,b}(\varepsilon, z) F_{c,b}(\varepsilon, z)^T).$$

For every $v \in \mathbb{R}^\ell$, the scalar $v^T H_\varepsilon v$ is the value of λ on a sum of squares of polynomials of degree at most b . Thus $H_\varepsilon \succeq 0$. The Taylor identity (5.15) implies that P_ε and H_ε have the same coefficients through degree b in ε . Here $\deg p \leq b$ ensures that the polynomial patch is retained exactly, and $\lambda(1) = 1$ retains the constant term of the affine pencil. Both tested expressions are polynomials in ε , so

$$\|P_\varepsilon - H_\varepsilon\| = O(\varepsilon^{b+1}), \quad P_\varepsilon \succeq -O(\varepsilon^{b+1}) I_\ell. \quad (5.17)$$

To turn this approximate positivity into exact feasibility, let

$$K_0 = \bigcap_{\mathcal{L}(x,u) \succeq 0} \ker \mathcal{L}(x, u), \quad S_0 = K_0^\perp.$$

Every selected matrix $\mathcal{L}(p(s), u(s))$ annihilates K_0 . Its Taylor coefficients at c therefore do so as well. Since P_ε is obtained by applying λ to its Taylor polynomial through degree b , we have

$$P_\varepsilon K_0 = 0. \quad (5.18)$$

In particular, any negative eigenvalues of P_ε occur on S_0 .

There is a feasible pencil matrix $P_* = \mathcal{L}(x_*, u_*)$ whose restriction to S_0 is positive definite. Indeed, choose a feasible matrix of maximum rank. For any other feasible matrix P , the average $(P_* + P)/2$ is feasible, and $\ker(P_* + P) = \ker P_* \cap \ker P$. Maximality of the rank therefore forces $\ker P_* \subset \ker P$, proving $\ker P_* = K_0$. If $S_0 = 0$, equation (5.18) already gives $P_\varepsilon = 0$, so $x_\varepsilon \in C$ and there is nothing to prove.

Otherwise choose $\mu > 0$ with $P_*|_{S_0} \succeq \mu I_{S_0}$ and put $\delta_\varepsilon = \varepsilon^b$. For sufficiently small positive ε ,

$$((1 - \delta_\varepsilon)P_\varepsilon + \delta_\varepsilon P_*)|_{S_0} \succeq (\mu \varepsilon^b - C_1 \varepsilon^{b+1}) I_{S_0} \succeq 0.$$

Both summands annihilate K_0 , so the mixed matrix is positive semidefinite on all of $\mathbb{R}^\ell$. By affineness of the pencil it is the pencil value at

$$\tilde{x}_\varepsilon = (1 - \delta_\varepsilon)x_\varepsilon + \delta_\varepsilon x_*, \quad \tilde{u}_\varepsilon = (1 - \delta_\varepsilon)u_\varepsilon + \delta_\varepsilon u_*.$$

Thus $\tilde{x}_\varepsilon \in C$. The vector x_ε is polynomial in ε and hence bounded near zero, giving (5.16). $\square$

5.3 The contradiction

Theorem 5.6. *For the admissible quadratic matrix map Q of (4.7), the set E_Q is not a spectrahedral shadow. Consequently its associated entire closed hyperbolicity cone K_Q is not a spectrahedral shadow.*

Proof. Suppose that E_Q has a finite affine semidefinite lift. Take the polynomial patch x , the functional Λ , and the open ball Ω from Lemmas 5.1 and 5.2. By Lemma 5.3, extend Λ through degree 32 so that it is positive definite on squares of polynomials of degree at most $b = 16$. This extension leaves all tested patch coordinates unchanged, since their degrees are at most four, and preserves $\Lambda(1) = 1$.

Apply Lemma 5.5 with $C = E_Q$, $p = x$, and $b = 16$. Fix a center c in the resulting nonempty open subset of Ω , and abbreviate the tested data by $x_\varepsilon = (A_\varepsilon, t_\varepsilon, y_\varepsilon)$. There are feasible points

$$\tilde{x}_\varepsilon = (A_\varepsilon + \Delta A_\varepsilon, t_\varepsilon + \Delta t_\varepsilon, y_\varepsilon + \Delta y_\varepsilon) \in E_Q$$

with all three perturbations of norm $O(\varepsilon^{16})$. By (5.5), $A_\varepsilon \succeq a_c \varepsilon^4 I_r$. Therefore, for sufficiently small $\varepsilon > 0$,

$$A_\varepsilon + \Delta A_\varepsilon \succeq \frac{1}{2} a_c \varepsilon^4 I_r, \quad \|A_\varepsilon^{-1}\| + \|(A_\varepsilon + \Delta A_\varepsilon)^{-1}\| = O(\varepsilon^{-4}).$$

The inverse identity gives

$$\begin{aligned} (A_\varepsilon + \Delta A_\varepsilon)^{-1} - A_\varepsilon^{-1} \\ = -(A_\varepsilon + \Delta A_\varepsilon)^{-1} \Delta A_\varepsilon A_\varepsilon^{-1} = O(\varepsilon^8) \end{aligned}$$

in operator norm. The vectors y_ε are bounded, and Q is quadratic, so

$$Q(y_\varepsilon) = O(1), \quad Q(y_\varepsilon + \Delta y_\varepsilon) - Q(y_\varepsilon) = O(\varepsilon^{16}).$$

Combining these estimates and using that r is fixed yields

$$\begin{aligned} t_\varepsilon + \Delta t_\varepsilon - \text{tr}((A_\varepsilon + \Delta A_\varepsilon)^{-1} Q(y_\varepsilon + \Delta y_\varepsilon)) \\ = t_\varepsilon - \text{tr}(A_\varepsilon^{-1} Q(y_\varepsilon)) + O(\varepsilon^8) \\ = -\kappa_c \varepsilon^4 + o(\varepsilon^4) < 0 \end{aligned}$$

for all sufficiently small positive ε . Since $A_\varepsilon + \Delta A_\varepsilon$ is positive definite, this contradicts the trace inequality defining the feasible point $\tilde{x}_\varepsilon \in E_Q$. Thus E_Q has no finite semidefinite lift. Proposition 2.2 identifies E_Q as a projection of an affine section of K_Q , so the same conclusion holds for K_Q . $\square$

Proof of Theorem 1.1. The construction in Section 4 gives finite dimensions $m = 330$ and $r = 20N - 1$ and a quadratic matrix map Q satisfying $Q(y) \succeq 0$ for every $y \in \mathbb{R}^m$. Proposition 2.1 shows that its polynomial p_Q is hyperbolic with respect to the specified direction. Theorem 5.6 shows that the entire closed hyperbolicity cone K_Q has no finite semidefinite lift. These are all the assertions of Theorem 1.1. $\square$

A A rank certificate for the Hankel seed

We prove Lemma 3.1. Recall that

$$H = \mathbb{R}[x_0, \dots, x_7]_2 = L \oplus P, \quad \dim H = 36, \quad \dim L = 8, \quad \dim P = 28,$$

where L consists of the quadratic monomials divisible by x_0 . We seek a positive semidefinite Hankel matrix T of rank 20 whose restriction to L is positive definite. Its projected kernel $K = \text{proj}_P(\ker T)$ must have independent pairwise products and span all pure cubics after multiplication by pure linear forms. The remaining requirement is

$$\dim((\ker T) \cdot H_0) = 321, \quad H_0 = \{p \in H : T(p, L) = 0\},$$

inside the 330-dimensional quartic space.

Our starting matrix is the moment matrix of twenty points arranged in two cyclic orbits. At that matrix the last product space has dimension 302. We exhibit a tangent direction for which nineteen independent product directions appear to first order. All nonvanishing assertions are certified below by explicit minors over $\mathbb{F}_7$; we then explain why these certificates produce a real positive semidefinite matrix with the required properties.

A.1 Cyclic orbit matrices and the required rank increase

For the algebraic construction work first over $\mathbb{C}$, keeping x_0 as the homogenizing coordinate. Give the eight variables the respective weights

$$\omega = (0, 5, 1, 9, 2, 8, 4, 6) \pmod{10}. \quad (\text{A.1})$$

Let ζ be a primitive tenth root of unity. For $a = (1, a_1, \dots, a_7)$, take the two orbits of $(1, \dots, 1)$ and a under the diagonal action $x_j \mapsto \zeta^{\omega_j} x_j$. Give every orbit point weight $1/10$, and let $T(a)$ be the resulting quadratic moment matrix. If m is a quartic monomial, its moment is

$$t_a(m) = \begin{cases} 1 + m(a), & \text{wt}(m) = 0, \\ 0, & \text{wt}(m) \neq 0. \end{cases} \quad (\text{A.2})$$

In particular $T(a)_{u,v} = t_a(uv)$ for quadratic monomials u, v . Formula (A.2) has integer coefficients in the parameters and also defines the matrices used in the finite-field calculation; no tenth roots of unity in $\mathbb{F}_7$ are needed.

Write I_i for the ordered quadratic monomials of weight i and set

$$C_i(a) = \begin{pmatrix} 1 & \cdots & 1 \\ u_1(a) & \cdots & u_{|I_i|}(a) \end{pmatrix}, \quad (u_1, \dots, u_{|I_i|}) = I_i.$$

The only nonzero block in column weight i of $T(a)$ is $C_{-i}(a)^T C_i(a)$. If every $C_i(a)$ has rank two, then $\text{rank } T(a) = 20$ and its kernel is the direct sum of the nullspaces $J_i = \ker C_i(a)$. Thus $J = \ker T(a)$ has dimension 16.

Let D denote a matrix whose columns are the quartic products of a basis of J with a basis of H_0 . For a Hankel variation T_1 satisfying

$$T_1|_{J \times J} = 0, \quad (\text{A.3})$$

write D_1 for the first-order product variation obtained by differentiating the equations for bases of J and H_0 . We give explicit formulas below and later realize T_1 as the velocity of a smooth arc. We shall establish, simultaneously with the two product conditions on K ,

$$\text{rank } D = 302, \quad \text{rank} \begin{pmatrix} D & 0 \\ D_1 & D \end{pmatrix} \geq 623 = 2 \cdot 302 + 19. \quad (\text{A.4})$$

These ranks do not depend on the choices of differentiable bases. Indeed, a change of the derivatives of those bases changes D_1 by a matrix of the form $AD + DB$, which is removed by block row and column operations in the second matrix of (A.4).

We first record the upper bounds that give the target ranks their meaning. They apply in characteristic zero wherever T has rank 20, $J \cap L = 0$, and K spans all pure cubics after multiplication by pure linear forms. The seven translations and the dilation in the chart $x_0 = 1$ have independent derivatives on the kernel graph, by the proof of Lemma 3.3, which uses only these hypotheses and works over $\mathbb{C}$ as well as over $\mathbb{R}$. Recall the obstruction to a dependence involving dilation: it would allow a translation making the kernel equal to the homogeneous space K . Cubic spanning would then annihilate all chart moments of degrees three and four, forcing rank at most $8 + 1 = 9$ instead of 20.

A translation or dilation preserves the projected kernel K . Continue each kernel vector with its highest homogeneous part fixed, so that its first derivative lies in L . Differentiating the kernel equation therefore gives $T'(0)J \subset T(L)$, so its quartic functional annihilates JH_0 . Scaling T gives one further independent annihilator, since scaling has zero kernel derivative whereas the eight affine motions have independent kernel derivatives. Hence, throughout the rank-20 locus under consideration,

$$\text{rank } D \leq 330 - 9 = 321. \quad (\text{A.5})$$

At an orbit matrix, scaling belongs to the twenty-dimensional family obtained by varying the individual weights of its twenty points. Rank 20 means that their evaluation functionals are independent already on quadratics, and hence on quartics by multiplication by x_0^2 . Each such quartic evaluation annihilates JH_0 , and all weight variations have zero kernel derivative. The eight affine motions remain independent modulo these twenty directions. Thus at the orbit matrix

$$\text{rank } D \leq 330 - 20 - 8 = 302. \quad (\text{A.6})$$

In particular, attaining rank at least 302 there is an open condition equivalent to equality.

A.2 Bases and first derivatives modulo seven

All entries in the remainder of this subsection and the next are in $\mathbb{F}_7$. Use the specialization

$$a = (1, 6, 6, 0, 3, 0, 2, 5). \quad (\text{A.7})$$

Index monomials by nondecreasing strings of variable indices, ordered lexicographically, both globally and within each weight. For example, 23 means x_2x_3 and 0011 means $x_0^2x_1^2$. Every row and column number below starts at one within its indicated block.

For C_i , take pivot columns

$$E_i = (1, 2) \quad \text{except that} \quad E_0 = E_8 = (1, 3). \quad (\text{A.8})$$

Use the nullspace basis having an identity matrix on the remaining coordinates, in their order. The quadratic blocks and resulting columns of J_i are as follows. A digit string in the third column records one entire column vector in the order of I_i .

i	I_i	columns of J_i
0	00, 11, 23, 45, 67	61000, 00610, 40201
1	02, 17, 34	421
2	04, 22, 56, 77	4210, 2401
3	15, 24, 36	601
4	06, 13, 44, 57	6010, 0601
5	01, 26, 37	511
6	07, 12, 46, 55	4210, 2401
7	14, 27, 35	151
8	05, 33, 47, 66	6100, 3031
9	03, 16, 25	601

Each specified two-column pivot is invertible. Thus every C_i has rank two and T has rank 20. The block on L is also invertible: in the order 00, 01, $\dots$, 07 its determinant is 6.

For later calculations, the ordered list of weight-zero quartics is

$$\begin{aligned}
&0000, 0011, 0023, 0045, 0067, 0126, 0137, 0225, 0334, 0447, \\
&0466, 0556, 0577, 1111, 1123, 1145, 1167, 1244, 1257, 1346, \\
&1355, 2233, 2247, 2266, 2345, 2367, 3356, 3377, 4446, 4455, \\
&4567, 4777, 5557, 5666, 6677.
\end{aligned} \tag{A.9}$$

Define T_1 by setting its quartic moments outside weight zero equal to zero and its moments in the order (A.9) equal to

$$(2, 0, 0, 0, 3, 6, 1, 4, 1, 6, 0, 3, 0, 5, 2, 2, 0, 0, 4, 0, 6, \underbrace{0, \dots, 0}_{14}). \tag{A.10}$$

The weight-zero block of products in $\text{Sym}^2 J$ has seventeen columns. Its transpose times the vector (A.10) is the zero vector. Products of every other weight pair trivially with T_1 , so (A.3) holds.

We specify a basis N_i for the weight- i part of H_0 . There is one coordinate of L in each weight except 3 and 7. Denote its within-block index by L_i ; thus $L_0 = 1$ corresponds to 00. If weight $-i$ has no L coordinate, let N_i be the identity matrix. Otherwise let N_i be the nullspace basis of the single row T_{L-i, I_i} , using pivot L_i and identity on the free indices. Its pivot is nonzero because $T|_L$ is invertible.

The following formulas specify derivatives of both bases. Put J'_i equal to zero off the pivot rows E_i , and set

$$(J'_i)_{E_i} = -(T_{E-i, E_i})^{-1} (T_1)_{E-i, I_i} J_i. \tag{A.11}$$

If N_i is an identity matrix, put $N'_i = 0$. In the other cases, put N'_i equal to zero off row L_i , and set

$$(N'_i)_{L_i} = -(T_{L-i, L_i})^{-1} (T_1)_{L-i, I_i} N_i. \tag{A.12}$$

These are the differentiated kernel equations with the free coordinates held fixed. For J'_i , tangency ensures that solving the two selected equations also solves all the equations $TJ'_i + T_1J_i = 0$: the image of T_1J_i lies in the image of T , and the two selected rows give coordinates on that image in the relevant weight block. The formulas for N'_i follow by differentiating $T(L, N_i) = 0$.

A.3 The explicit minors

We now give the minors that establish the two product conditions on K and the rank increase in (A.4). All product matrices use the following convention. In weight w , multiply a column u from block i by a column v from block $w - i$, adding

$$u_{jj'}v_{kk'} \quad \text{to row} \quad \text{sort}(jj'kk').$$

Order columns lexicographically by the weight i , the column number of u , and the column number of v . For products from $\text{Sym}^2 J$, retain only pairs for which the first index (i , column number) is at most the second. Thus each unordered product occurs once, with its ordinary polynomial coefficients.

Let G_1 be this matrix for $J \cdot J$, restricted to quartic rows with no zero index. It is the product matrix for $\text{Sym}^2 K$. Let G_2 be the product matrix of J with the seven coordinate vectors x_0x_j , $1 \leq j \leq 7$, restricted to rows with exactly one zero index. Removing that zero identifies it with the product map from K times pure linear forms to pure cubics. Rows and columns are numbered anew after the restriction to a weight and to these row sets. In all the tables, a colon denotes the inclusive range.

The following square minors use all columns of G_1 and all rows of G_2 , respectively. The specified omissions determine the other index set completely.

G_1				G_2			
w	omit rows	size	det.	omit columns	size	det.	
0	15, 19:22	17	3	8, 10, 11	8	6	
1	9, 11, 12, 15:20	11	4	9:11	8	6	
2	13, 15, 19:22	16	4	8, 10, 11	9	2	
3	3, 10, 14:20	11	5	8, 10	8	6	
4	13, 15, 18, 19, 21, 22	16	6	10:12	9	6	
5	9, 11, 13, 15:20	11	5	8, 10, 11	8	4	

The number of columns of G_1 in weights $0, \dots, 5$ is 17, 11, 16, 11, 16, 11; the number of rows of G_2 is 8, 8, 9, 8, 9, 8. Thus these determinants give precisely the full column and full row ranks needed in these weights.

For D , use first factors from J and second factors from H_0 , with the preceding product order. Form D_1 by replacing the first factor by its derivative in (A.11), replacing the second by its derivative in (A.12), and adding the two resulting product columns. In weights $0, \dots, 5$, the dimensions of D are respectively

$$35 \times 48, \quad 31 \times 42, \quad 35 \times 47, \quad 31 \times 43, \quad 35 \times 47, \quad 31 \times 42.$$

The next table specifies a square submatrix $E = D[R, S]$ in each weight by listing the omitted rows and columns; R, S always denote the retained indices.

w	omit rows	omit columns	size	det E
0	13, 34, 35	5, 9, 11, 33, 34, 36:38, 40:43, 45:48	32	6
1	24, 26, 31	7, 9, 10, 28, 30, 32, 33, 36:42	28	1
2	32, 34, 35	12, 14:16, 18, 19, 35, 38, 40, 41, 43:47	32	6
3	26, 31	14, 16, 17, 19, 20, 35:43	29	3
4	33, 34, 35	17, 20, 21, 23:25, 27, 28, 41:47	32	5
5	19, 26, 31	17, 18, 20, 22, 23, 25, 26, 36:42	28	3

To obtain the bordered minors, use the indices R, S in both diagonal positions of $\begin{pmatrix} D & 0 \\ D_1 & D \end{pmatrix}$. Add rows R' in its second block row and columns S' in its first block column. Set $A = D[R, S']$ and $B = D[R', S]$. Eliminating the two copies of E leaves the corner

$$Z = D_1[R', S'] - D_1[R', S]E^{-1}A - BE^{-1}D_1[R, S'] + BE^{-1}D_1[R, S]E^{-1}A. \quad (\text{A.13})$$

The corner matrices are the following, with rows separated by semicolons when their entries are listed.

w	R'	S'	Z	$\det Z$
0	34	38	(5)	5
1	24, 31	38, 40	(4, 2; 2, 4)	5
2	34, 35	45, 46	(6, 4; 2, 0)	6
3	26, 31	37, 39	(3, 6; 0, 5)	1
4	33, 35	43, 46	(6, 4; 1, 2)	1
5	26, 31	37, 38	(2, 6; 2, 0)	2

In particular the resulting bordered minor has size $2|R| + 1$ in weight zero and $2|R| + 2$ in each of weights one through five. The bordered determinant is, up to its row and column ordering sign, $(\det E)^2 \det Z$, as follows directly by eliminating the two diagonal copies of E . Thus the table proves the asserted bordered lower bounds without requiring an upper bound on the rank of D in the finite field.

For completeness, every entry in these tables can be checked with the following fixed elimination rule. At column n , interchange row n with the first row at or below it whose entry t in that column is nonzero. Divide the new row by t and subtract the appropriate multiples from every other row, always reducing modulo seven. The determinant is the product of the pivots, with one minus sign for each interchange. Appended columns give the inverse actions in (A.13) by the same rule. For the G_2 minor in weight zero, the signed pivots are

$$(-2, -2, -5, -1, 4, -1, 1, 5).$$

For E in weight zero they are

$$\begin{aligned} &1, 4, 1, 4, -4, -1, -4, -5, -5, -4, -4, 6, -4, 4, 6, -3, \\ &-3, -1, 6, 6, 4, 5, -5, 2, 1, 5, 2, 5, 6, 4, 4, 4. \end{aligned} \quad (\text{A.14})$$

Together with the monomial order, the basis rules, and the moment vectors above, this specifies the finite calculation using matrices with at most thirty-five quartic rows in each weight.

A.4 Characteristic zero and simultaneous real data

We now pass from the explicit modular minors to actual orbit matrices and tangent directions in characteristic zero. First lift (A.7) to the same integer vector. The chosen pivots of the C_i , the entries used as pivots for N_i , and the 2×2 blocks in (A.11) are all invertible modulo seven. Hence the corresponding bases and inverse formulas are defined over $\mathbb{Z}_{(7)}$, the rational numbers whose denominators are not divisible by seven.

There is one point to address before lifting the bordered minors: the first variation must remain tangent exactly, not just modulo seven. The weight-zero part of $\text{Sym}^2 J$ has dimension seventeen.

Its seventeen product columns are independent, since their restriction to the pure rows contains the nonzero 17×17 minor of G_1 above. Consequently tangency of a weight-zero functional gives seventeen independent equations on its thirty-five moment coordinates. Choose as pivot coordinates the seventeen quartic rows of that minor. Lift the remaining eighteen coordinates of (A.10) to integers and solve these seventeen equations over $\mathbb{Z}_{(7)}$. The solution reduces to precisely the displayed moment vector. It therefore defines an exact characteristic-zero tangent whose basis derivatives reduce to (A.11) and (A.12). Every nonzero determinant in the preceding tables thus proves that its defining rational function is nonzero in characteristic zero.

It remains to obtain all weights and a real positive semidefinite matrix simultaneously. We spell out the genericity argument to make clear that the tangent need not be chosen separately for different weights. The permutation

$$(x_2, x_3), \quad (x_4, x_5), \quad (x_6, x_7) \quad \text{interchanged, with } x_0, x_1 \text{ fixed,}$$

negates every weight and preserves the family of orbit matrices. The nonzero minors for weights 1, 2, 3, 4 therefore also give nonempty rank conditions for weights 9, 8, 7, 6, respectively.

More formally, take the Zariski open subset of the parameter space $\mathbb{C}^7$ on which all C_i have rank two, $T|_L$ is invertible, and the seventeen weight-zero products are independent. This is a nonempty irreducible parameter space. The solutions of the seventeen tangency equations form a rank-eighteen algebraic vector bundle over it: locally an invertible seventeen-column minor solves for seventeen moment coordinates in terms of the other eighteen. Its total space is irreducible. Each condition certified above is a nonempty Zariski open subset of this space, as is its image under the weight-negating permutation, after restricting to the common nonempty parameter domain. Their finite intersection is nonempty. The same reasoning for the minors of G_1 and G_2 , which depend only on a , gives both product conditions on K .

The resulting lower bounds in weights $0, \dots, 9$ are

w	0	1	2	3	4	5	6	7	8	9	sum
rank G_1	17	11	16	11	16	11	16	11	16	11	136
rank G_2	8	8	9	8	9	8	9	8	9	8	84
rank D	32	28	32	29	32	28	32	29	32	28	302
rank $\begin{pmatrix} D & 0 \\ D_1 & D \end{pmatrix}$	65	58	66	60	66	58	66	60	66	58	623

(A.15)

Here $136 = \dim \text{Sym}^2 K$ and 84 is the dimension of the pure cubic space. The first two rows prove the required injectivity and surjectivity. Equation (A.6) then turns the third row into equality, and the fourth row gives (A.4).

Impose now the real structure in which x_0, x_1 are real and the pairs (x_2, x_3) , (x_4, x_5) , (x_6, x_7) are conjugate. Its fixed parameter set

$$a_1 \in \mathbb{R}, \quad a_3 = \overline{a_2}, \quad a_5 = \overline{a_4}, \quad a_7 = \overline{a_6}$$

is a real form of $\mathbb{C}^7$ and is Zariski dense. One way to see this is to write each conjugate pair as $(u + iv, u - iv)$: this is an invertible complex linear change of coordinates from a real coordinate space, and a polynomial vanishing on that real space is zero. The weight-zero tangent equations at a fixed real parameter are preserved by the real structure. Their solution space is therefore the complexification of its real fixed subspace, which is likewise Zariski dense. A nonempty open condition in the tangent bundle is attained over a nonempty open set of parameters, and within each such fiber it is an open condition on a linear space. We can thus choose simultaneously a parameter in the real form and a real tangent satisfying every rank condition in (A.15).

For these parameters all twenty orbit points are real in the corresponding real coordinates. Indeed multiplication by ζ^{ω_j} respects the conjugate pairs and multiplies the weight-five coordinate by ± 1 . Since every point has positive weight $1/10$, the moment matrix is real positive semidefinite. Its rank is 20, and its nonsingular restriction to L is positive definite. Choose ordinary real coordinates on this fixed real form, keeping x_0 and the pure-variable span, and relabel them $x_0, \dots, x_7$. The induced polynomial change of coordinates preserves $L \oplus P$ and multiplication, and hence the total product and first-variation ranks above. Thus these are real Hankel data in the conventions of Section 3.

A.5 Deformation to the seed matrix

We have obtained a real positive semidefinite orbit matrix T and a real Hankel tangent T_1 satisfying (A.4). Independence of the products in $\text{Sym}^2 K$ implies independence of those in $\text{Sym}^2 J$ by taking highest chart degree. Lemma 3.2, or its Schur-complement proof, therefore shows that the rank-20 Hankel locus is smooth near T , with tangent condition (A.3). There is a real smooth arc $T(t)$ on this locus with $T(0) = T$ and $T'(0) = T_1$. For sufficiently small t , its twenty nonzero eigenvalues remain positive, its restriction to L remains positive definite, and the two product conditions on $K(t)$ remain valid.

Choose smooth local kernel and H_0 bases along the arc and let $D(t)$ be their product matrix. Apply fixed invertible row and column operations that put $D(0)$ in rank normal form

$$D(0) = \begin{pmatrix} I_{302} & 0 \\ 0 & 0 \end{pmatrix}.$$

Writing the transformed derivative in the same blocks as $\begin{pmatrix} A & B \\ C & F \end{pmatrix}$, elementary block elimination gives

$$\text{rank} \begin{pmatrix} D(0) & 0 \\ D'(0) & D(0) \end{pmatrix} = 604 + \text{rank } F.$$

Thus $\text{rank } F \geq 19$. Eliminate the invertible leading 302×302 block of $D(t)$. Its remaining Schur complement is $tF + o(t)$, since both off-diagonal blocks vanish at $t = 0$. A nonzero 19×19 minor of F remains nonzero after division by t^{19} for every sufficiently small nonzero t . Hence $\text{rank } D(t) \geq 321$ there. The upper bound (A.5) applies along the arc and gives equality. Taking one such $T(t)$ proves Lemma 3.1.

Reproducible calculation. The accompanying source file `verify_certificate.py` reconstructs the monomial blocks, bases, differentiated products, and every displayed minor using exact integer arithmetic modulo seven. It also verifies the ranks in weights six through nine at the very same specialization (A.7) and tangent (A.10). This supplies an independent reproduction of the finite calculation; the explicit minors and the characteristic-zero argument above establish the lemma without relying on the program.

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